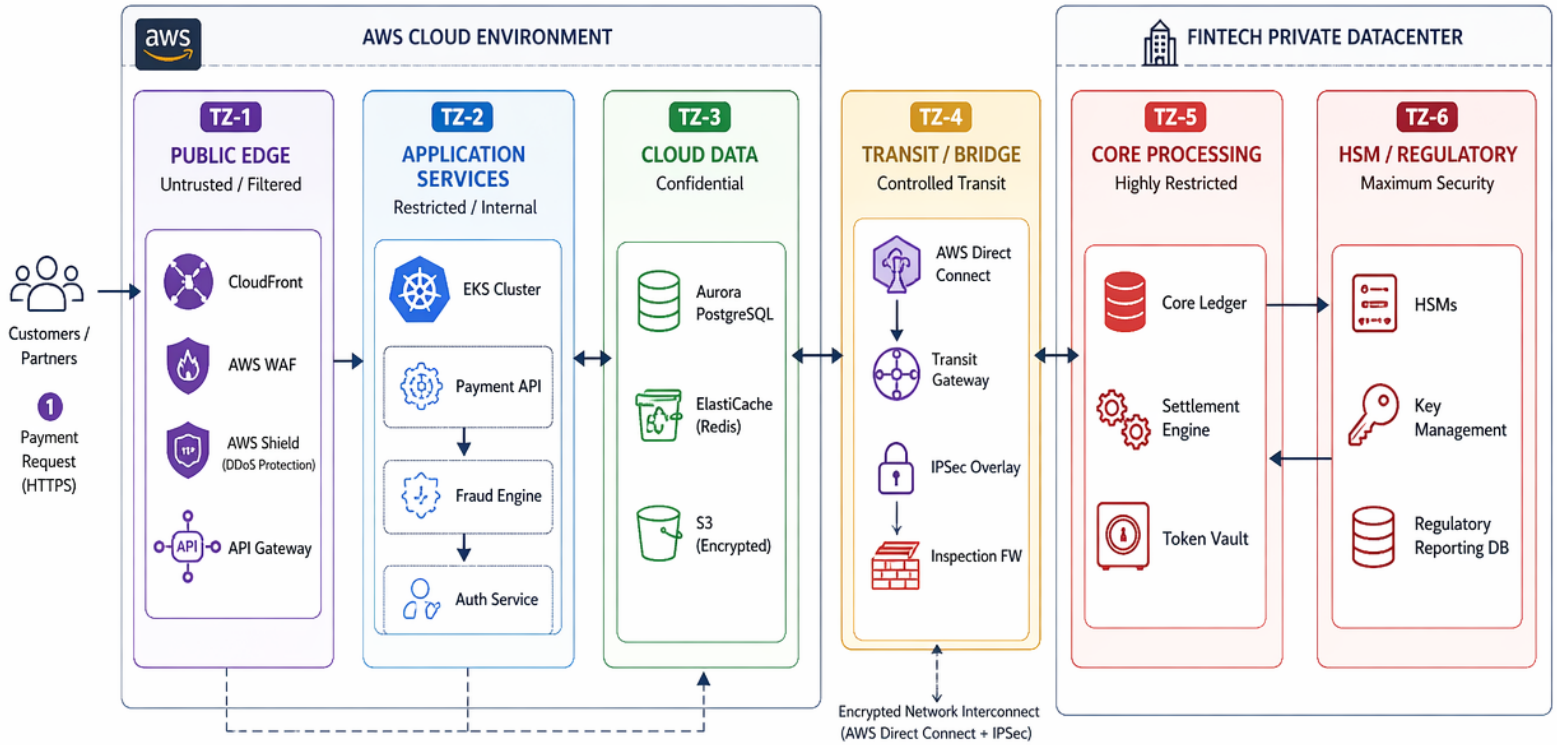
FINTECH HYBRID CLOUD REFERENCE ARCHITECTURE

Trust Zone Model and Data Flow



TRUST ZONE PRINCIPLES

- Six explicit trust zones
- No implicit trust between zones
- All cross-zone traffic is authenticated, authorized, encrypted and logged
- Least privilege and explicit policies enforced at every boundary

TRUST ZONE LEGEND

- TZ-1 Public Edge (Untrusted / Filtered)
- TZ-2 Application Services (Restricted)
- TZ-3 Cloud Data (Confidential)
- TZ-4 Transit / Bridge (Controlled)
- TZ-5 Core Processing (Highly Restricted)
- TZ-6 HSM / Regulatory (Maximum Security)

→ Primary Data Flow

- - → Service / Control Flow

Authenticated, Encrypted Cross-Zone Traffic

CONTROLLED DATA FLOW (EXAMPLE: CUSTOMER PAYMENT)

1. Customer payment request enters at TZ-1. WAF and API Gateway perform initial validation.
2. Routed to TZ-2. Payment API authenticates request, invokes Fraud Engine for risk scoring.
3. Application data retrieved from TZ-3 (databases, cache, objects) as needed.
4. If core ledger posting or tokenization is required, request crosses TZ-4 transit bridge (authenticated, encrypted, inspected).
5. Processed in TZ-5 by Core Ledger / Settlement Engine / Token Vault.
6. Cryptographic operations delegated from TZ-5 to TZ-6 (HSM / KMS).
7. Response returns back through the same controlled path to TZ-2 and then to TZ-1.

CROSS-ZONE ENFORCEMENT (APPLIES AT EVERY BOUNDARY)

- Authentication (Strong identity)
- Authorization (Least privilege)
- Encryption (In transit & at rest)
- Logging & Audit (Immutable logs)
- Inspection & Monitoring (Threat detection)
- Policy Enforcement (Explicit allow rules only)